

KSP Crime Intelligence Platform — System Architecture

Conversational AI + Crime Analytics, built directly on the official Police FIR database schema

OverviewUsersLayersData SchemaAPI EndpointsModulesChatbot / AICatalyst ServicesSecurity

Platform Overview

00

What we're building, why, and what makes it different

The Problem

FIR records sit in disconnected station-level systems. No AI, no cross-district intelligence, no proactive policing. Bond tracking is a physical register. Criminal pursuit relies on walkie-talkies. Everything is manual and reactive.

Our Solution

An intelligent conversational AI + crime analytics platform, built directly on top of the real CaseMaster-centric FIR schema. Officers query records in natural language (English + Kannada + voice), surface hidden criminal networks, predict crime before it happens, and run daily operations from one screen.

Our Edge

Real insights from serving police experience. Good Conduct Bond tracker, rowdy/gang monitoring, cross-district escape alerts, court case tracker, festival-window crime prediction — all grounded in actual FIR/CrimeNo structure, not guesswork.

Challenge

Challenge 2

Conversational AI + Crime Analytics Platform — all modules

Deployment

100% Zoho Catalyst

18 Catalyst services, mandatory stack

Frontend

React Web Dashboard

Browser-based, no install, any station, any screen

Data Grounding

Real FIR Schema

CaseMaster, Accused, Victim, ArrestSurrender, Act/Section — exact column names used end-to-end

Users & Access Model

01

One login · One dashboard · Full access · Verified police officers only



Inspector

Morning dashboard · Court case tracker · Bond expiry alerts · Station ops overview · All analytics



Constable

AI chatbot queries · FIR lookup · Crime response assistant · PDF reports · Voice input



SCRB Analyst

Statewide analytics · Network graphs · Predictive dashboards · Trend reports



Station Clerk

Complainant/victim/accused entry · Duty roster · Event log · Complaint intake

Auth Model — Simple, Secure, Practical

Login via Employee.KGID (Karnataka Government ID) + Password through Catalyst Authentication → session token → full dashboard unlocked. No UI role separation — inspector can direct any constable to run any query. Every action silently logged to an immutable audit table, keyed to KGID, queryable by supervisors.

02

System Architecture — 8 Layers

Full top-down flow, browser to data storage

LAYER 1 — USERSPolice officers, browser-based, no install, works on any device

Inspector

Station head — morning ops, court tracking, alerts

Constable

Chatbot queries, FIR lookup, crime response, PDF export

SCRB Analyst

State-level analytics, predictions, network analysis

Station Clerk

FIR data entry — complainant, victim, accused, events



LAYER 2 — FRONTENDReact · Catalyst AppSail · D3.js · Leaflet.js · Recharts · TailwindCSS

AI Chatbot UI

Voice input · English + Kannada toggle · Evidence trail panel · PDF export

Crime Hotspot Map

Leaflet.js plotting CaseMaster.latitude/longitude · district drill-down

Criminal Network Graph

D3.js force graph — Accused, Victim, Location nodes, shared-case edges

Inspector Dashboard

Today's FIRs, court hearings, bond expiry, crime spike alerts

Offender Profile

Full Accused history across CaseMaster, ArrestSurrender, risk score

Court Case Tracker

Hearings, ChargesheetDetails, verdict history

Analytics Dashboard

Trend charts by CrimeHead/CrimeSubHead, seasonal patterns

Clerk Entry Form

New FIR → complainant/victim/accused/act-section capture



LAYER 3 — SECURITY GATEWAYCatalyst API Gateway · HTTPS/TLS 1.3 · rate limiting · request validation

TLS Termination

All traffic encrypted in transit, enforced at gateway

Rate Limiting

Per-officer throttling to prevent abuse/scraping

Schema Validation

Every payload validated against CaseMaster/Accused/Victim field types before reaching Functions



LAYER 4 — AUTHCatalyst Authentication · KGID + Password · session tokens

Login

Employee.KGID + password → session token issued

Session Validation

Every Function call re-validates token before touching DataStore



LAYER 5 — APPLICATION (CATALYST FUNCTIONS)Node.js/Python serverless functions — one group per module, all schema-aware

Case Functions

CrimeNo/CaseNo generation, CaseMaster CRUD, status transitions

Person Functions

Complainant, Victim, Accused, ArrestSurrender writes

Legal Functions

Act/Section lookups, ActSectionAssociation linking

Analytics Functions

Aggregations, hotspot clustering, network graph builders



LAYER 6 – AI / MLQuickML (LLM + RAG + Text-to-SQL) · Zia AutoML · Zia Voice/NLP

Query Classifier

Routes officer question → Text-to-SQL (structured) or RAG (unstructured case notes)

Text-to-SQL

Prompted with exact CaseMaster/Accused/Victim/Act/Section column names — zero hallucinated fields

Zia AutoML

Recidivism risk scoring, hotspot prediction, gang-pattern flagging



LAYER 7 – DATACatalyst DataStore (relational, FIR schema) · NoSQL (notes, chat, embeddings) · Cache

DataStore

CaseMaster, Accused, Victim, ComplainantDetails, ArrestSurrender, Act, Section, org hierarchy — exact ER schema

NoSQL

Free-text case notes, chat history, RAG vector embeddings, audit log

Cache

Hot reads — dashboard counts, lookup tables (Act, District, Rank)



LAYER 8 – INFRA / OPSCron · Signals · Catalyst Mail · SmartBrowz · Stratus · Pipelines

Cron

Bond expiry checks, nightly gang-pattern scans

Signals

Real-time alerts — escape routing, threshold breaches

SmartBrowz

Watermarked PDF generation for reports/chargesheets

Pipelines

CI/CD deployment of Functions + AppSail build

Every table below is created exactly as-is in DataStore. Column names are never renamed — Text-to-SQL prompts and SmartBrowz templates reference them directly.

●Core Transactional Tables		
TABLE	KEY COLUMNS	PURPOSE
CaseMaster	CaseMasterID PK · CrimeNo · CaseNo	Root FIR record — Cat+District+Station+Year+Serial encoded in CrimeNo, lat/long, incident dates, all category/status/court FKs
ComplainantDetails	ComplainantID PK · CaseMasterID FK	Complainant identity + demographics (occupation, religion, caste)
Victim	VictimMasterID PK · CaseMasterID FK	Victim records, multiple per case, VictimPolice flag
Accused	AccusedMasterID PK · CaseMasterID FK	Accused records, PersonID sorting (A1, A2...)
ArrestSurrender	ArrestSurrenderID PK · CaseMasterID FK · AccusedMasterID FK	Arrest/surrender events, IO, court, jurisdiction
inv_arrestsurrenderaccused	junction	One arrest event ↔ multiple accused
ActSectionAssociation	CaseMasterID FK · ActID FK · SectionID FK	Many-to-many: which act-sections apply to which case
ChargesheetDetails	CSID PK · CaseMasterID FK	Chargesheet date + outcome type (A=Chargesheet, B=False Case, C=Undetected)

●Legal Reference Tables		
TABLE	KEY COLUMNS	PURPOSE
Act	ActCode PK	Legal acts (IPC, NDPS...)
Section	ActCode FK · SectionCode	Sections under each act
CrimeHead	CrimeHeadID PK	Major crime classification group
CrimeSubHead	CrimeSubHeadID PK · CrimeHeadID FK	Minor crime classification (e.g. Murder, Robbery)
CrimeHeadActSection	CrimeHeadID FK · ActCode FK · SectionCode FK	Maps crime heads to relevant act-sections

●Org & Geography Hierarchy		
TABLE	KEY COLUMNS	PURPOSE
State	StateID PK	Top of geography hierarchy
District	DistrictID PK · StateID FK	State → District
Unit	UnitID PK · TypeID FK · ParentUnit (self-ref)	Police stations / circle offices, self-referencing hierarchy
UnitType	UnitTypeID PK	Station vs circle vs district office, with Hierarchy level
Employee	EmployeeID PK · KGID · UnitID FK · RankID FK · DesignationID FK	Officer master — KGID used as login identifier
Rank / Designation	RankID / DesignationID PK	Officer rank & role lookups
Court	CourtID PK · DistrictID FK	Court master

●Lookup / Master Tables	
TABLE	PURPOSE

CaseCategory	FIR / UDR / PAR / Zero FIR
GravityOffence	Heinous / Non-Heinous
CaseStatusMaster	Under Investigation / Charge Sheeted / Closed
CasteMaster / ReligionMaster / OccupationMaster	Complainant demographic lookups

Open item: the relationship matrix references a one-to-one Inv_OccuranceTime table off CaseMaster, but its columns aren't defined anywhere — the likely fields (IncidentFromDate, IncidentToDate, InfoReceivedPSDate, latitude, longitude) already live directly on CaseMaster. Confirm with the data source whether this is a separate table or redundant before building it.

REST API — Catalyst Functions

04 All endpoints prefixed /server/ksp-api/ · require Authorization: Bearer <session_token> · every call audit-logged with KGID + timestamp

● Case / FIR			
METHOD	ENDPOINT	MAPS TO	DESCRIPTION
POST	/cases	CaseMaster	Register new FIR — server generates CrimeNo & CaseNo
GET	/cases/{caseMasterId}	CaseMaster + joins	Full case view: complainant, victims, accused, sections, status, court
GET	/cases?stationId=&categoryId=&statusId=&dateFrom=&dateTo=	CaseMaster	Filtered case search
PATCH	/cases/{caseMasterId}/status	CaseMaster.CaseStatusID	Update case status
GET	/cases/{caseMasterId}/timeline	derived	FIR → Investigation → Chargesheet → Court → Verdict → Bond

● Complainant / Victim / Accused			
METHOD	ENDPOINT	MAPS TO	
POST	/cases/{id}/complainants	ComplainantDetails	
POST	/cases/{id}/victims	Victim	
POST	/cases/{id}/accused	Accused (auto-assigns PersonID A1, A2...)	
GET	/accused/{accusedMasterId}/profile	Accused + ArrestSurrender, cross-case lookup	

● Acts & Sections			
METHOD	ENDPOINT	MAPS TO	
GET	/acts	Act (Active=1)	
GET	/acts/{actCode}/sections	Section where ActCode	
POST	/cases/{id}/sections	ActSectionAssociation	
GET	/crime-heads/{id}/act-sections	CrimeHeadActSection	

● Arrest / Surrender / Court			
METHOD	ENDPOINT	MAPS TO	
POST	/arrests	ArrestSurrender + inv_arrestsurrenderaccused	

GET	/arrests?accusedId=	Cross-case arrest history for one accused
GET	/arrests?districtId=&stateId=	Jurisdiction arrest activity — feeds escape-alert routing
POST	/chargesheets	ChargesheetDetails
GET	/courts/{courtId}/hearings	CaseMaster + ArrestSurrender where CourtID

●Org / Geography (reference)

METHOD	ENDPOINT	MAPS TO
GET	/units/{unitId}	Unit + UnitType + District + State
GET	/units/{unitId}/children	Unit where ParentUnit = unitId
GET	/employees/{employeeId}	Employee + Rank + Designation + Unit
GET	/districts/{districtId}/stats	Aggregated CaseMaster counts by district

●AI / Chatbot

METHOD	ENDPOINT	BACKED BY
POST	/chat/query	Query classifier → Text-to-SQL or RAG
POST	/chat/voice	Zia Speech-to-Text → /chat/query
GET	/chat/{queryId}/evidence	Evidence trail: source CaseMasterIDs, fields, confidence
POST	/chat/export-pdf	SmartBrowz → watermarked PDF

●Analytics / Prediction

METHOD	ENDPOINT	BACKED BY
GET	/analytics/hotspots?districtId=	CaseMaster lat/long clustering
GET	/analytics/risk-score/{accusedMasterId}	Zia AutoML, trained on Accused + ArrestSurrender + CaseMaster history
GET	/analytics/gang-network?accusedMasterId=	Graph from shared cases / MO across Accused & Victim
POST	/alerts/escape	Route prediction → Signals fan-out to stations on route

●Bond / HS Record Tracker EXTENSION

METHOD	ENDPOINT	MAPS TO
POST	/bonds	BondRecord (new table: CaseMasterID FK, AccusedMasterID FK, signDate, expiryDate, status)
GET	/bonds/expiring?withinDays=30	Cron-driven → Signals alert

Functional Modules

05

What the platform actually does, grouped by capability

M01FIR Registration & Case Management

- Structured CrimeNo generation: 1-digit category + 4-digit district + 4-digit station + 4-digit year + 5-digit serial, unique per station/category/year
- Complainant, victim, accused entry forms linked by CaseMasterID
- Act/Section tagging via searchable Act → Section picker
- Full case timeline view from registration to verdict

M02 Offender Tracking

- Cross-case Accused profile — every FIR an individual appears in, across districts
- ArrestSurrender history with IO, court, jurisdiction
- AI risk scoring via Zia AutoML on arrest/case history
- Good Conduct Bond / HS Record Tracker **INNOVATION** — Cron alerts 30/60/90 days before expiry, one-click renewal PDF
- Gang membership tagging — link accused across organized crime groups

M03 Investigator Decision Support

- Automated case summary generation via QuickML LLM, grounded in CaseMaster.BriefFacts
- Similar-case retrieval via RAG — matching CrimeMajorHeadID/CrimeMinorHeadID + location
- Case Lifecycle Tracker **INNOVATION** — FIR → Investigation → Chargesheet → Court → Verdict → Bond on one screen
- One-click court hearing PDF package via SmartBrowz

M04 Crime Forecasting & Early Warning

- Nightly Zia AutoML run on CaseMaster history — emerging pattern detection per district
- Predictive hotspot risk score updated daily, shown on map
- Cross-District Escape Alert **INNOVATION** — officer flags accused escaping → route predicted → stations on route alerted via Signals + Mail
- Gang activity flag — same MO across 3+ FIRs in 7 days triggers organized-crime tag

M05 Explainable AI & Governance

- Every chatbot answer has an expandable Evidence Trail: source CaseMasterIDs, matched fields, confidence
- All responses grounded in exact DataStore records — no hallucination, no invented fields
- Immutable audit log: KGID, full query, full answer, sources, timestamp
- PII masking on bulk export — complainant/victim names masked unless authorized

06

Chatbot & NLP Architecture

One conversational interface, grounded entirely in the real schema

1. Input

Text or voice (Zia Speech-to-Text). English or Kannada (Zia NLP). Officer types or speaks: "show all accused linked to FIR 1044300062026..."

2. Query Classifier

LLM decides: structured lookup (Text-to-SQL against CaseMaster/Accused/Victim/Act/Section) vs. unstructured retrieval (RAG over case notes)

3. Execution

Text-to-SQL prompt is few-shot grounded in the exact table/column list — never invents a field that doesn't exist in the schema

4. Response

Answer composed in plain language, with optional table/chart, returned to chat UI

5. Evidence Trail

Expandable panel: which CaseMasterIDs/fields were used, confidence score, last-updated timestamp

6. Audit

KGID + full query + full answer + sources logged immutably to NoSQL audit collection

07

Zoho Catalyst Services Used

18 services, 100% Catalyst stack

DataStore

Relational FIR schema

NoSQL

Notes, chat, embeddings, audit log

Authentication

KGID + password login

API Gateway

TLS, rate limiting, routing

Functions

Serverless application logic

QuickML

LLM, RAG, Text-to-SQL

Zia AutoML

Risk scoring, forecasting

Zia Voice/NLP

Speech-to-text, Kannada

Signals

Real-time alerts

Cron

Scheduled jobs — bond expiry, nightly scans

Cache

Hot reads, lookup tables

Catalyst Mail

Notifications, alerts

SmartBrowz

Watermarked PDF generation

Stratus

File/object storage

Pipelines

CI/CD deployment

AppSail

React frontend hosting

08

Security & Compliance

Schema-grounded, end to end

Authentication & Audit

Catalyst Auth via Employee.KGID + password. Every API call logged: KGID, endpoint, params, timestamp → immutable NoSQL audit log, queryable by supervisors.

Encryption

DataStore encrypted at rest. TLS 1.3 enforced in transit via API Gateway.

PII Masking

ComplainantDetails.ComplainantName and Victim.VictimName masked in bulk exports unless explicitly authorized.

Watermarking

Every SmartBrowz PDF stamped with generating officer's KGID + timestamp for traceability.

Sensitive Case Flagging

POCSO / sensitive cases flagged via CaseCategoryID or GravityOffenceID — restricted export, visual marker.

Containment

No crime data ever leaves the Catalyst environment — all AI processing (QuickML, Zia) runs on-platform.